ScienceX

Battery Storage Market Notes, October 2026 issue

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The price of electricity changes with the time of day. "Grid-scale batteries" (large batteries connected to the transmission grid that store electricity and release it) earn money from that difference. These notes use published figures and our own records to show this month's movements in figures and tables.

1. The three ways a battery storage plant earns money (how to read these notes)

A battery storage plant (grid-scale battery) earns income from three markets. ① Wholesale electricity (JEPX) is the market where you buy and store electricity when it is cheap and sell it when it is expensive. ② The balancing market (EPRX) is where you sell grid operators the "capacity" to fix surpluses and shortages of electricity. ③ The capacity market (OCCTO) is where the country buys future electricity supply capacity in advance. These notes present the figures for the three markets in that order. The ①, ② or ③ at the start of each section title shows which market it covers.

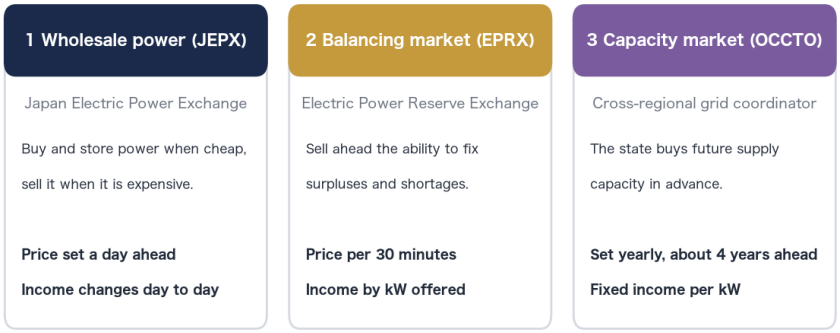


Figure 1 The three ways a battery storage plant earns money

Source: market rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

2. Three movements this month

We picked three movements this month from the figures for the three markets.

- In wholesale electricity (JEPX), the monthly average spread between the highest 4 hours and the lowest 4 hours in October 2026 was 17.14 yen/kWh in Tokyo and 19.30 yen/kWh in Kyushu.
- In the balancing market (EPRX), the bid-to-requirement ratio for Primary reserve passed 100% for the first time in June 2026, at 106.8%. It was 123.1% in September 2026, and the average clearing price was 3.43 yen/kW per 30 min.
- In the balancing market (EPRX), 56.9% of Tertiary reserve 2 revenue in September 2026 came from the top 10% of 30-minute slots by revenue.

3. ① Wholesale electricity (JEPX) | Average price

Wholesale electricity is the electricity that power companies and retailers buy and sell. Its price is set on the "spot market" of the Japan Electric Power Exchange (JEPX) (a market where electricity for the next day is traded the day before). "System (nationwide)" is the price when all of Japan is treated as one market (the system price). When the transmission lines between regions are congested, the price splits by area, from Hokkaido to Kyushu (area prices).

The average wholesale price for System (nationwide) in October 2026 was 22.51 yen/kWh, against 11.67 yen/kWh in the same month last year.

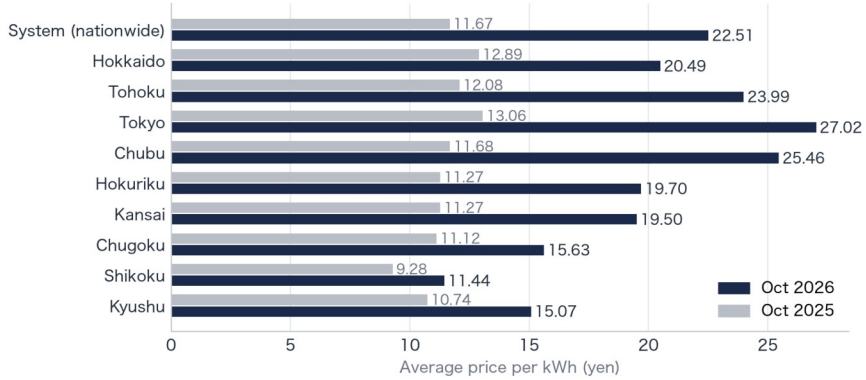


Figure 2 Average wholesale electricity price (yen/kWh). Navy: October 2026; grey: same month last year

Source: spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of 30-minute prices. Our tally.

4. ① Wholesale electricity (JEPX) | How the price moves through the day

The wholesale price changes every 30 minutes. The figure shows the shape of an average day: the price at each time of day, averaged across the month. If a battery charges when the price is low and discharges when it is high, the difference is its income.

The daily average low and high for wholesale electricity were 19.03 around 12:00 and 37.11 around 16:30 in Tokyo, and 4.94 around 12:00 and 24.74 around 17:30 in Kyushu (yen/kWh).

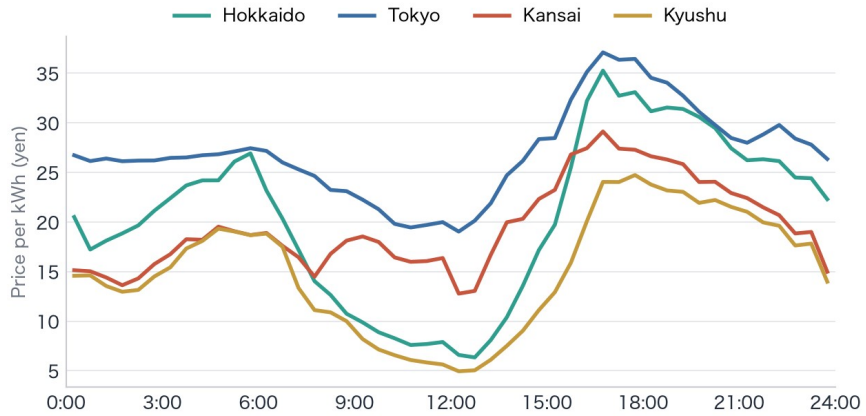


Figure 3 Daily highs and lows of the wholesale price (October 2026 average, yen/kWh)

Source: JEPX spot market. Our tally.

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5. ① Wholesale electricity (JEPX) | How large is the spread per kWh

For each area, we took the average price of the day's highest 4 hours, subtracted the average of the lowest 4 hours, and averaged the result over the month. This is the difference per kWh if you bought 1 kWh in the cheapest 4 hours and sold it in the most expensive 4 hours. It is a simple difference: energy lost in charging and discharging and fees are not deducted, and the order of the cheap and expensive hours is ignored.

By area, the monthly average spread between the highest 4 hours and the lowest 4 hours was largest in Hokkaido at 29.02 yen/kWh and smallest in Tokyo at 17.14 yen/kWh.

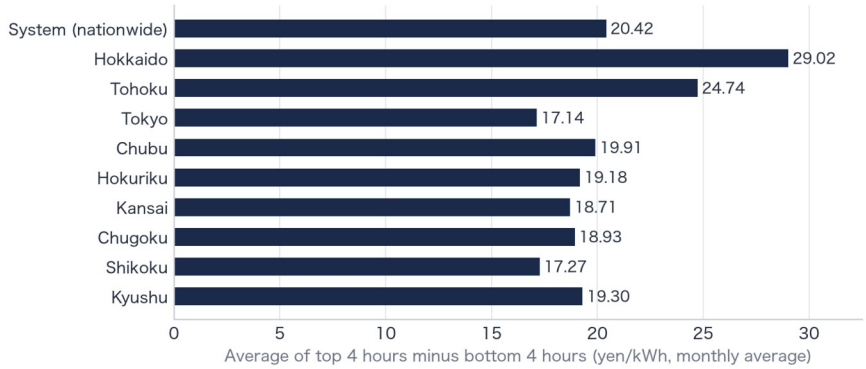


Figure 4 Wholesale spread: average of the highest 4 hours − average of the lowest 4 hours (October 2026, yen/kWh, monthly average)

Source: JEPX spot market. Our tally.

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	Days with a 4-hour spread of 10 yen or more
System (nationwide)	22.95	20.42	29
Hokkaido	32.56	29.02	31
Tohoku	27.09	24.74	30
Tokyo	19.62	17.14	29
Chubu	22.37	19.91	29
Hokuriku	21.71	19.18	31
Kansai	21.22	18.71	31
Chugoku	20.90	18.93	31
Shikoku	19.40	17.27	28
Kyushu	21.19	19.30	31

The spread is the monthly average of the daily spreads. October 2026 covers 31 days.

6. ① Wholesale electricity (JEPX) | There is a spread every day

We show the same 4-hour spread day by day. The bigger the spread on a given day, the more room a battery has to earn from it.

Days with a 4-hour wholesale spread of 10 yen or more were 29 of 31 for System (nationwide), 31 in Hokkaido, Hokuriku, Kansai, Chugoku and Kyushu, and 28 in Shikoku.

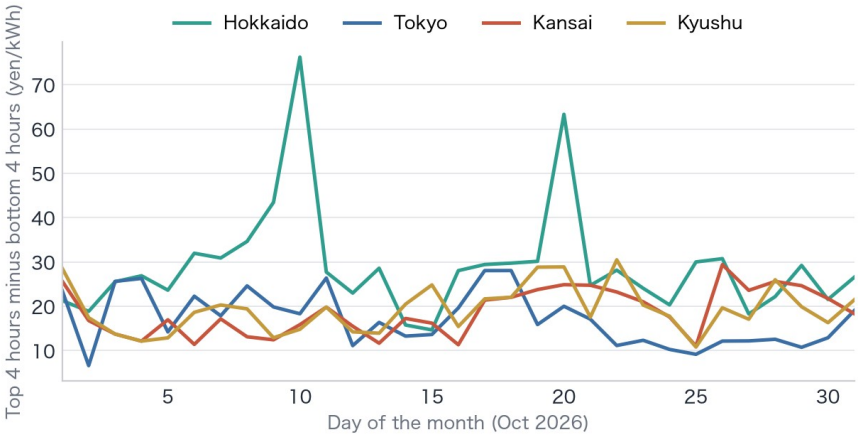


Figure 5 Daily wholesale spread (average of the day's highest 4 hours − average of its lowest 4 hours, yen/kWh)
Source: JEPX spot market. Our tally.

7. ② Balancing market (EPRX) | Six products, and prices by area

The balancing market is where grid operators buy, every 30 minutes, the "capacity" to fix surpluses and shortages of electricity. Products differ by how quickly output must change after an instruction. Primary reserve is the fastest and Tertiary reserve 2 the slowest. A Composite product is a bid that offers several products together. Prices are set in yen per 1 kW of capacity per 30 minutes. The figure shows the average clearing price and the clearing rate (the share of bid volume that was awarded) by area (by the area where the generating resource is located). Darker colors mean higher values. Nationwide figures are in the table in the "Market figures" section. Published figures for October 2026 are not yet in our tally, so we show September 2026 (preliminary), the latest month with figures for all products.

By the area where the resource is located, the highest average clearing price in the balancing market in September 2026 was Primary reserve in Tohoku at 8.18 yen, and the lowest was Tertiary reserve 2 in Shikoku at 0.38 yen (yen/kW per 30 min).

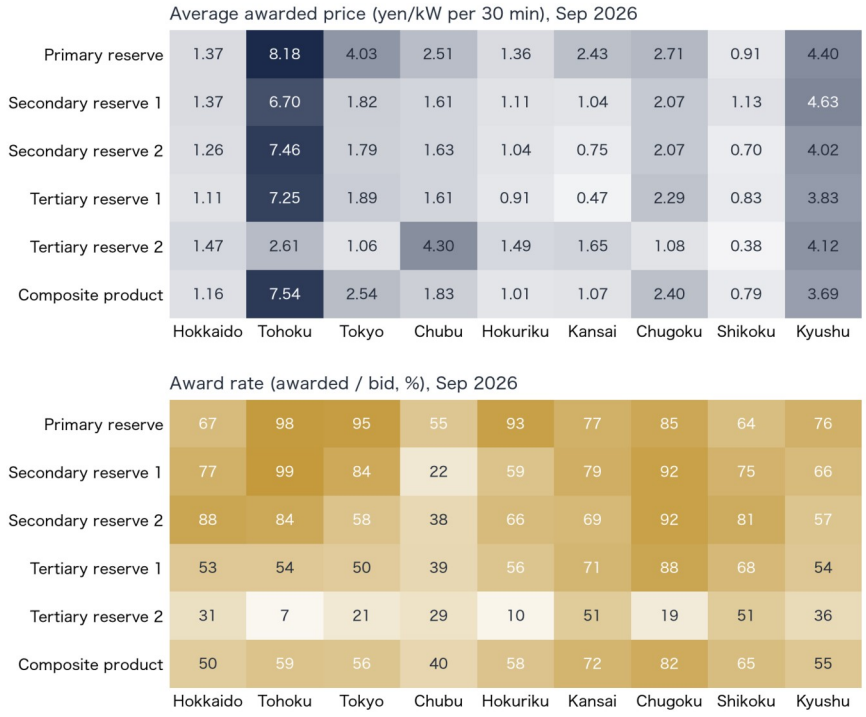


Figure 6 Balancing market average clearing price (top) and clearing rate (bottom) by area (September 2026, by area where the resource is located)

Source: trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

Product	Time from instruction to output change	Duration
Primary reserve	Within 10 seconds	5 minutes or more
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes

8. ② Balancing market (EPRX) | Primary reserve became a contest for places

Primary reserve is the product that batteries win most. The bars show the bid-to-requirement ratio (bid volume ÷ volume requested by grid operators). Above 100%, more bids come in than were requested, and some bids are rejected. The line shows the average clearing price. Monthly figures from April 2025.

The bid-to-requirement ratio for Primary reserve in the balancing market passed 100% for the first time in June 2026 (106.8%), and was 123.1% in September. The average clearing price was 3.43 yen/kW per 30 min, against 3.57 yen in the same month last year.

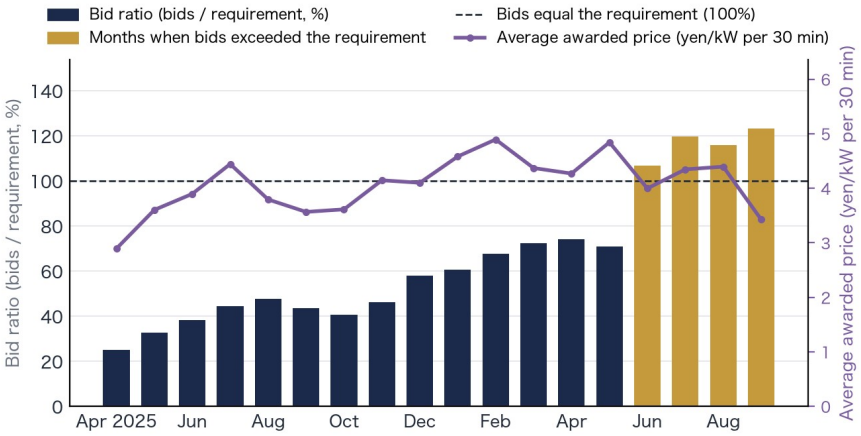


Figure 7 Primary reserve bid-to-requirement ratio (bars) and average clearing price (line) (monthly; gold bars are months when bids exceeded the requirement)

Source: trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

9. ② Balancing market (EPRX) | How prices have moved by product

Average clearing price by product since April 2025 (monthly average weighted by awarded volume). From March 14, 2026, Primary reserve through Tertiary reserve 1 and the Composite product moved from 3-hour trading to 30-minute trading (Tertiary reserve 2 was already traded in 30-minute slots).

The average clearing price for Primary reserve in the balancing market was 2.89 yen in April 2025 and 3.43 yen in September 2026, with a high of 4.90 yen in February 2026 (yen/kW per 30 min).

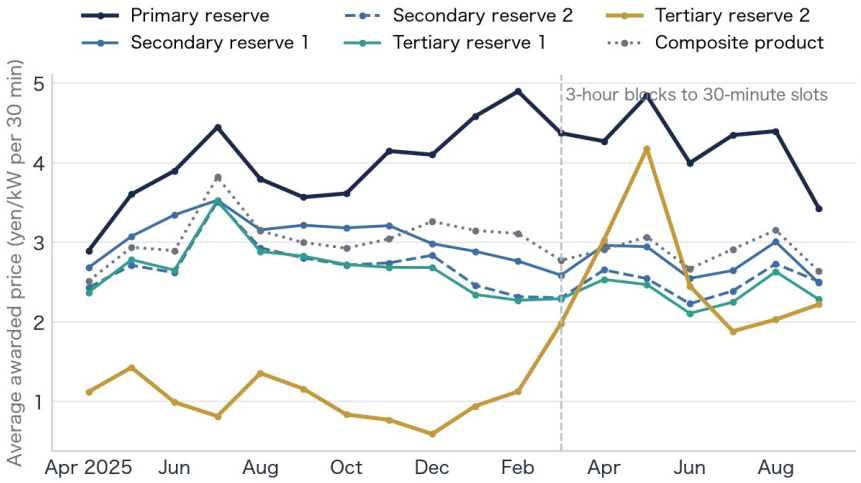


Figure 8 Balancing market average clearing price by product (yen/kW per 30 min, monthly)

Source: trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

10. ② Balancing market (EPRX) | Which hours of the day are expensive

Average clearing price by product for each hour of the day in September 2026.

For Primary reserve in the balancing market in September 2026, the average clearing price was highest in the 18:00 hour at 4.09 yen and lowest in the 0:00 hour at 2.75 yen (yen/kW per 30 min).

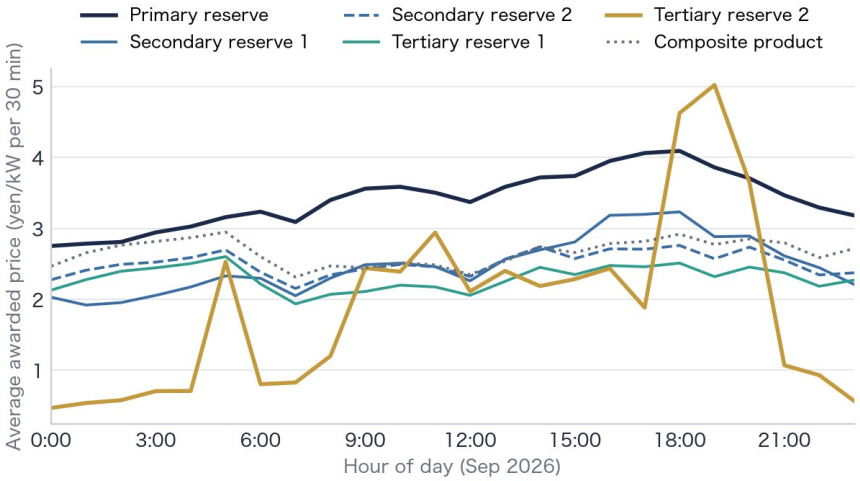


Figure 9 Balancing market average clearing price by time of day (September 2026, yen/kW per 30 min)

Source: trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

11. ② Balancing market (EPRX) | Which products batteries are used for

The figure shows the share of awarded volume held by batteries. The table compares the average clearing price of the volume batteries won with the average clearing price for the product as a whole.

For Primary reserve in the balancing market (September 2026), batteries held a 27.9% share. The average clearing price of the volume batteries won was 5.17 yen/kW per 30 min, against 3.43 yen overall.

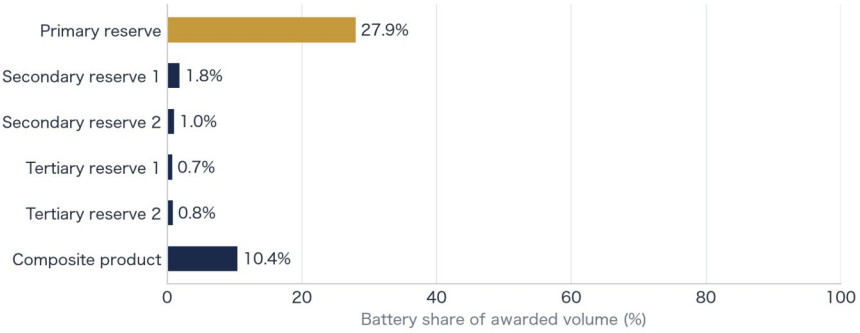


Figure 10 Share of batteries in the balancing market (%)

Source: trading results (preliminary) by resource type published by the Electric Power Reserve Exchange (EPRX). Our tally.

Product	Won by batteries (yen/kW per 30 min)	Product overall (yen/kW per 30 min)
Primary reserve	5.17	3.43
Secondary reserve 1	2.18	2.49
Secondary reserve 2	1.85	2.51
Tertiary reserve 1	1.86	2.28
Tertiary reserve 2	59.26	2.22
Composite product	5.13	2.64

Both are averages weighted by awarded volume (September 2026, preliminary, nationwide).

12. ② Balancing market (EPRX) | Revenue is concentrated in a small number of slots

For each product, we ranked the month's 30-minute slots by revenue (average clearing price × awarded volume, nationwide) and plotted what percentage of the month's revenue the top slots account for. The more the curve bulges above the dotted line, the more the revenue is concentrated in a few slots. Tertiary reserve 2 is a product that covers errors in forecasting renewable output caused by the weather. Its price is low in most slots and high in a small number of them.

In the balancing market (September 2026), the share of revenue in the top 10% of slots was 56.9% for Tertiary reserve 2 and 15.8% for Primary reserve.

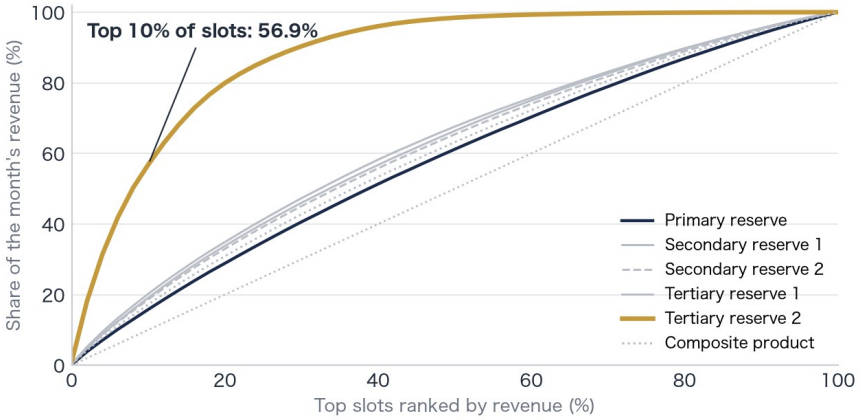


Figure 11 Share of balancing market revenue in the top slots (September 2026, nationwide)

Source: trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

In September 2026, of the 1,194 slots with awards, the top 1% by revenue (average clearing price × awarded volume) accounted for 9.4% of the month's revenue. There were 25 slots with a nationwide average clearing price of 10 yen/kW per 30 min or more, and 198 slots with no awards.

Slots with the highest maximum clearing price (by area where the resource is located):

Date	Time	Area	Maximum clearing price (yen/kW per 30 min)
September 8	9:30	Tokyo	96.50
September 8	10:30	Tokyo	96.50
September 8	11:00	Tokyo	96.50
September 27	11:00	Tokyo	96.00
September 27	11:30	Tokyo	96.00

Source: Tertiary reserve 2 trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our tally.

13. ③ Capacity market (OCCTO) | A market that buys future supply capacity

The capacity market buys electricity supply capacity 4 years ahead through a once-a-year auction. The price is set in yen per kW per year. The left figure shows clearing prices in past main auctions. The right figure shows the battery and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (a separate auction that supports new builds such as batteries and pumped hydro). These figures are published once a year, so they do not change every month. The latest is delivery FY2029 (published January 2026).

In the capacity market, the Tokyo clearing price went from 3,495 yen/kW for delivery FY2025 to 15,111 yen/kW for FY2029. In the FY2025 round of the Long-Term Decarbonization Power Source Auction, contracted battery and pumped hydro capacity was 819 MW of lithium-ion and 886 MW of non-lithium. Both are the latest figures, published once a year.

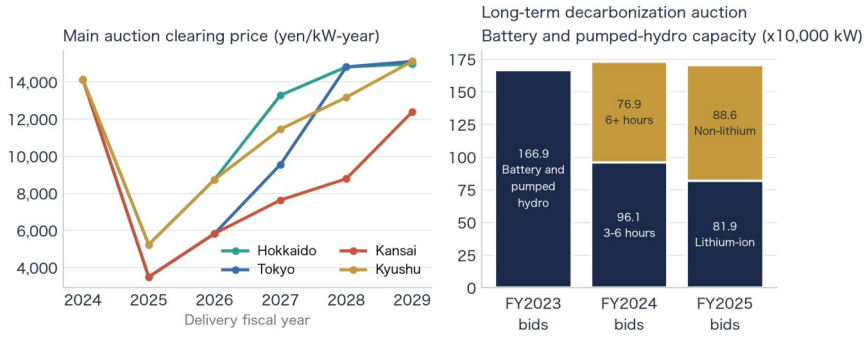


Figure 12 Capacity market clearing prices (left, yen/kW) and battery and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: clearing results and capacity securing contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in section 15). As published.

14. Our records | Where battery storage projects are increasing

We counted the projects newly listed this month on DealRoom (our data room for buyers, a site where we list battery storage projects), by grid area of the grid operator. These are figures for projects we handled, not counts for the whole market. Transfer of rights is a sale that hands over the rights to the land and the power; Completed-build handover is a sale that hands over the plant once it is built.

New listings on our DealRoom in October 2026 were 77 projects, 309.1 MW. By grid area: Tokyo 33 projects, 99.8 MW; Tohoku 8 projects, 73.4 MW; Chubu 11 projects, 69.9 MW.

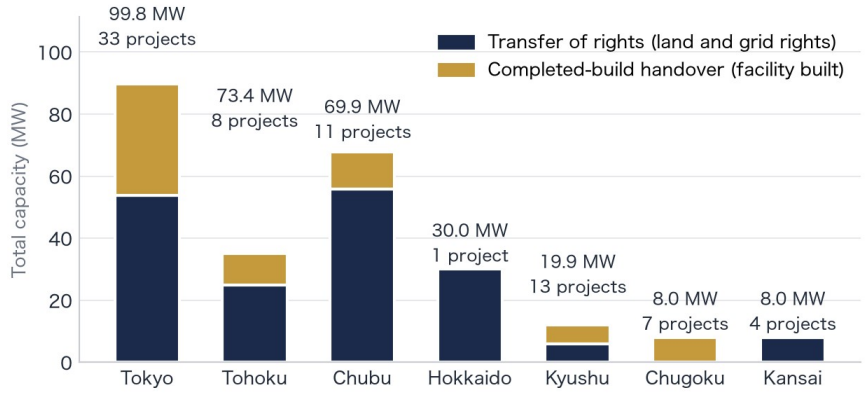


Figure 13 Output of projects listed on our DealRoom in October 2026 (MW)

Source: our DealRoom records (project locations shown only down to the grid area; prices and parties are not shown).

15. Our records | How many months and how much it takes to connect to the grid

A grid connection study is the procedure in which you ask the grid operator to check "Can we connect at this site, and how much will the work cost?" The grid connection cost contribution is the part of the connection work cost that the applicant pays. From the response letters we received, we tallied only high-voltage cases (connections to 6,600 V distribution lines) by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 cases are not shown.

The medians of grid connection study responses (high voltage only, as of October 19, 2026) ranged from 8.0 months in Tokyo to 17.0 months in Tohoku for construction time, and from ¥3.02 million in Tokyo to ¥32.35 million in Tohoku for the grid connection cost contribution.

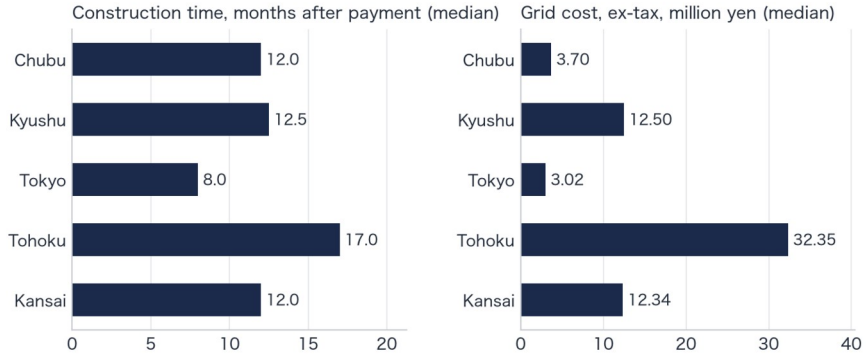


Figure 14 Median grid connection study responses (high voltage) by grid area (left: months of construction; right: grid connection cost contribution, excluding tax)

Source: tally of the high-voltage grid connection study response letters from general transmission and distribution operators that we received (as of October 19, 2026).

16. Market figures (tables you can use in your materials)

We put the figures behind the charts into tables you can use in your own materials: ① wholesale electricity (JEPX) figures and trading volumes by area, ② nationwide balancing market figures by product (bid-to-requirement ratio, clearing rate, price, battery share), and ③ capacity market clearing prices.

Wholesale electricity, JEPX spot market (October 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	4-hour spread	Maximum	Slots at 30 yen or more	Slots at 0.01 yen
System (nationwide)	22.51	11.67	16.02	31.06	20.42	47.88	220	39
Hokkaido	20.49	12.89	11.31	32.25	29.02	105.00	303	67
Tohoku	23.99	12.08	15.78	34.19	24.74	50.01	437	95
Tokyo	27.02	13.06	23.12	34.69	17.14	50.01	466	16
Chubu	25.46	11.68	20.38	33.55	19.91	46.45	387	24
Hokuriku	19.70	11.27	18.29	26.92	19.18	46.45	218	24
Kansai	19.50	11.27	18.22	26.76	18.71	46.45	215	29
Chugoku	15.63	11.12	9.86	23.10	18.93	44.79	63	79
Shikoku	11.44	9.28	7.50	16.61	17.27	37.51	23	146
Kyushu	15.07	10.74	8.31	23.10	19.30	44.79	53	91

Trading volume (October 2026, total for 31 days): total contracted volume 30,319 million kWh (same month last year 22,279 million kWh), sell bid volume 45,218 million kWh, buy bid volume 38,191 million kWh.

Source: spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Our tally.

Balancing market (September 2026, preliminary, nationwide)

Product	Requirement (MW)	Bid volume (MW)	Awarded volume (MW)	Bid-to-requirement ratio (bids ÷ requirement)	Clearing rate (awarded ÷ bids)	Average clearing price (yen/kW per 30 min)	Battery share
Primary reserve	1,388.7	1,710.0	1,337.3	123.1%	78.2%	3.43	27.9%
Secondary reserve 1	962.6	1,996.2	1,102.3	207.4%	55.2%	2.49	1.8%
Secondary	983.1	3,709.3	2,163.2	377.3%	58.3%	2.51	1.0%

Product	Requirement (MW)	Bid volume (MW)	Awarded volume (MW)	Bid-to-requirement ratio (bids ÷ requirement)	Clearing rate (awarded ÷ bids)	Average clearing price (yen/kW per 30 min)	Battery share
reserve 2							
Tertiary reserve 1	2,498.5	5,895.9	3,021.0	236.0%	51.2%	2.28	0.7%
Tertiary reserve 2	376.3	1,371.7	368.9	364.5%	26.9%	2.22	0.8%
Composite product	3,383.8	6,641.4	3,623.5	196.3%	54.6%	2.64	10.4%

Volumes are averages per 30-minute slot. A product with a bid-to-requirement ratio below 100% does not get enough bids to cover the requirement. The clearing rate is the share of bid volume that was awarded.

Source: trading results (preliminary) and trading results by resource type published by the Electric Power Reserve Exchange (EPRX). Our tally.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Battery and pumped hydro capacity contracted in the Long-Term Decarbonization Power Source Auction (10,000 kW)

Auction year	Category	Contracted capacity (10,000 kW)
FY2023 round	Batteries and pumped hydro	166.9
FY2024 round	3 to 6 hours	96.1
FY2024 round	6 hours or more	76.9
FY2025 round	Lithium-ion	81.9
FY2025 round	Non-lithium	88.6

Source: clearing results and capacity securing contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in section 15). As published.

17. Assumptions and terms

- Figures are tallied by us from the values published by the sources. We have not added up several months or converted anything to annual figures. We do not show revenue amounts for any single battery storage plant.
- Months covered: wholesale electricity (JEPX) is October 2026 (31 days); the balancing market (EPRX) is preliminary figures for September 2026. The balancing market covers a different month until the published figures for October 2026 are in our tally.
- Average prices are simple averages of 30-minute slots (not weighted by volume). Daytime is the 8:00–16:00 slots; evening is the 16:00–20:00 slots.
- Spread: the day's 30-minute slots are ranked by price, and the average of the cheapest 8 slots is subtracted from the average of the highest 8 slots (4 hours) (4 slots for 2 hours). The order of the cheap and expensive hours, energy lost in charging and discharging, and fees, taxes and transmission charges are not considered. This does not show actual revenue.
- Balancing market: nationwide figures (total column) from EPRX preliminary values. Requirement is by grid operator (TSO); bid volume, awarded volume and average clearing price are by the area where the resource is located. Volumes are averages per 30-minute slot. The average clearing price is weighted by awarded volume. Bid-to-requirement ratio = bid volume ÷ requirement; clearing rate = awarded volume ÷ bid volume. For the 3-hour blocks up to March 13, 2026, each block was counted as 6 slots of 30 minutes.
- The time from instruction to output change and the duration for each balancing market product are the FY2026 specifications.
- High-price slots for Tertiary reserve 2: slots with a nationwide average clearing price of 10 yen/kW per 30 min or more. The top 1% is the top 1% of slots ranked by revenue (average clearing price × awarded volume). The maximum clearing price is the published value by area where the resource is located.

Sources (OCCTO, capacity market clearing results): delivery FY2024 https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / delivery FY2025

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https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / delivery FY2026 https://www.occto.or.jp/iinkai/youryou/kentoukai/2022/files/youryou_kentoukai_43_03.pdf / delivery FY2027 https://www.occto.or.jp/market-board/market/oshirase/2023/files/240124_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2027.pdf / delivery FY2028 https://www.occto.or.jp/market-board/market/oshirase/2024/files/250129_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2028.pdf / delivery FY2029 https://www.occto.or.jp/assets/news/capacity-market/260120_mainauction_youryouyakujokekka_kouhyou_jitsujukyu2029.pdf

Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 round https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 round https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 round https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market clearing results for each delivery fiscal year are published in a different year (FY2024 in 2020, FY2029 in January 2026). The number of battery projects in the Long-Term Decarbonization Power Source Auction is not shown, because it is not in the published summary tables.

Sources: Japan Electric Power Exchange (JEPX) spot market trading results / Electric Power Reserve Exchange (EPRX) trading results (preliminary) and trading results by resource type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction clearing results and Long-Term Decarbonization Power Source Auction capacity securing contract documents / our tally of the grid connection study response letters we received / our DealRoom records.

Terms: system price (System (nationwide)) = the wholesale electricity price when all of Japan is treated as one market. Area price = the price that splits by area depending on congestion on the transmission lines. The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = a measure of power (1 MW = 1,000 kW). MWh = the amount of electricity that can be stored. yen/kWh = the price per 1 kWh of electricity.

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Newsletter "Japan grid-scale BESS: primary data": https://www.linkedin.com/newsletters/7521830530569445376/

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